ROLL NO: _____

MADHAV INSTITUTE OF TECHNOLOGY & SCIENCE, GWALIOR

(A Deemed to be University | Autonomous Institute Established in 1957 | NAAC 'A++' Accredited)

END-SEM EXAMINATION — DECEMBER / JUNE 2024

FIRST YEAR (COMMON TO ALL BRANCHES) – SEMESTER 2

Time Allowed: 3 Hours **Maximum Marks: 70**

NOTE: 1. Attempt any FIVE questions selecting at least ONE question from each Unit.
2. All questions carry equal marks. Sub-questions carry marks indicated in brackets [ ].
3. Assume suitable missing data, if any, and state it clearly.
4. Use of scientific non-programmable calculator is permitted.

--- UNIT I ---

Q.1 Explain Ecosystem Structure: Biotic and Abiotic components, Food Chains, Food Webs, Ecological Pyramids. [14 Marks]

[OR]

Q.1 (Alt) Describe Biogeochemical cycles: Carbon cycle, Nitrogen cycle, and Hydrological water cycle. [14 Marks]

--- UNIT II ---

Q.2 Explain Air Pollution: Primary and Secondary pollutants, Photochemical Smog, Acid Rain, Greenhouse Effect. [14 Marks]

[OR]

Q.2 (Alt) Describe Air pollution control devices: Cyclone Separator, Electrostatic Precipitator (ESP), Wet Scrubbers. [14 Marks]

--- UNIT III ---

Q.3 Explain Water Pollution: Dissolved Oxygen (DO), Biochemical Oxygen Demand (BOD), Chemical Oxygen Demand (COD). [14 Marks]

[OR]

Q.3 (Alt) Describe Wastewater Treatment stages: Primary sedimentation, Secondary activated sludge, and Tertiary disinfection. [14 Marks]

--- UNIT IV ---

Q.4 Explain Municipal Solid Waste Management (MSWM): 3R Concept (Reduce, Reuse, Recycle), Landfilling, Composting. [14 Marks]

[OR]

Q.4 (Alt) Describe E-Waste and Biomedical Waste hazards, handling rules, and incineration disposal methods. [14 Marks]

--- UNIT V ---

Q.5 Explain Disaster Management Cycle: Mitigation, Preparedness, Response, and Recovery. [14 Marks]

[OR]

Q.5 (Alt) Describe Natural Disasters (Earthquakes, Floods, Cyclones) and Industrial hazards (Bhopal Gas Tragedy case study). [14 Marks]

**** END OF QUESTION PAPER — MITS GWALIOR AUTONOMOUS EXAMINATION CELL ****